Roll No: _____

Thapar Institute of Engineering and Technology
Department of Computer Science and Engineering
Auxiliary Examination

BE CSE Third Year
Course Name: Quantum Computing
Time: 3 hours (5:30 am-8:30 pm)
Marks : 100

Course Code: UCS619
Date: August 27, 2025
Name of Faculty: Dr Samya Muhuri

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**Note: Attempt any ten questions. Assume missing data, if any, suitably. Give appropriate reasoning for your answers.**

1. Consider two-qubit quantum state

$$|\phi\rangle = \frac{\sqrt{2}}{3\sqrt{3}}|00\rangle - \frac{1}{\sqrt{6}}|01\rangle + \frac{2i\sqrt{2}}{3\sqrt{3}}|10\rangle - \frac{5i}{3\sqrt{6}}|11\rangle$$

   (a) What are the probabilities of outcome of 0 and 1 if first qubit of $|\phi\rangle$ is measured? **(3, CO1, BL3)**

   (b) What are the probabilities of outcome 0 and 1 if second qubit of $|\phi\rangle$ is measured? **(3, CO1, BL**

   (c) What is the state of the system after second qubit is measured as $|1\rangle$? **(4, CO1, BL3)**

2. Write down various steps of Shor's algorithm. Explain the procedure for factoring N=15. Please note that you need to explain the concept of Periodic function.

**(10, CO4, BL4)**

3. Given the search space $|000\rangle, |001\rangle, ..., |111\rangle$. Design the Grover's algorithm circuit (with the detailed oracle and Grover diffusion operator) to find the element 100 in the search space. Explain the value at each point for searching the element 100. **(10, CO4, BL3)**

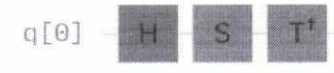
4. (a) The standard way to write the state of a qubit using the bloch sphere is

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

   show that on the y-axis of the Bloch Sphere $y_+ = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$ **(5, CO1, BL**

   (b) Examine the output when S-gate is operated on $|+\rangle$ and $|-\rangle$. **(5, CO2, BL2)**

5. (a) For the given circuit write the matrix equivalence. **(5, CO2, BL2)**




   (b) Prove that $X = HZH$ using matrix multiplication method. **(5, CO2, BL2)**

6. Alice creates a 8-bit string
$|0\rangle\,|1\rangle\,|+\rangle\,|0\rangle\,|0\rangle\,|-\rangle\,|+\rangle\,|-\rangle$
If Bob randomly measures in the following order using the bases
$\{|0\rangle, |1\rangle\}\{|\pm\rangle\}\{|0\rangle, |1\rangle\}\{|\pm\rangle\}\{|0\rangle, |1\rangle\}\{|0\rangle, |1\rangle\}\{|\pm\rangle\}\{|\pm\rangle\}$
describe the bit string that Alice and Bob keep using BB84 algorithm. **(10, CO4, BL3)**

7. (a) A quantum state is in the state

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{e^{i\theta}}{\sqrt{2}}|1\rangle$$

Express the state in the $|+\rangle$, $|-\rangle$ basis.

**(4, CO1, BL3)**

(b) Proof No cloning theorem with two methods. **(6, CO1, BL2)**

8. (a) Define Quantum Fourier Transform (QFT) for an n-qubit quantum system. **(4, CO4, BL1)**

(b) Let $n = 1$ qubit, $N = 2$ dimensional space. Apply QFT on the state $\alpha|0\rangle + \beta|1\rangle$. **(6, CO4, BL3)**

9 Construct a Quantum Circuit to implement the following four qubit Bell state $\frac{1}{\sqrt{2}}(|1001\rangle + |0110\rangle)$. Draw the quantum circuit and write down the QASM Code for the same. Give appropriate reasoning for your answers. **(10, CO2, BL4)**

10. Deutsch-Jozsa algorithm is used to check whether a function is constant or balanced. Draw the circuit of Deutsch's algorithm and explain the value at each point for the following scenario of three qubits: $f(000) = f(001) = f(100) = f(101) = 1$, $f(010) = f(011) = f(110) = f(111) = 0$.

**(10, CO4, BL4)**

11. Consider a Y error occurs at $2^{nd}$ qubit. Explain the status of the quantum system at each step of Shor's 9-qubit code. **(10, CO3, BL3)**

*Checked answer scripts will be shown on 29th August 2025 (Friday) from 2:00-2:30 p.m. at cabin L434, CSED block*

Roll No: _____

# Thapar Institute of Engineering and Technology
## Department of Computer Science and Engineering
**Special Auxiliary Examination**

**Course Name: Quantum Computing (CSE)** | **Course Code: UCS619**
Time: 3 hours | Date: Sept. 25, 2025
Max Marks: 100 | Name of Faculty: Dr. Vaibhav Pandey

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Note: Assume missing data, if any, suitably. Give appropriate reasoning for your answers.

1. Write down the unitary matrix for the following quantum gates:
   - Hadamard gate
   - T gate
   - S gate
   - Y gate
   - Z gate

**(2*5=10)**

2. A three-qubit system is in the state

$$|\psi\rangle = \left(\frac{\sqrt{2}+i}{\sqrt{20}}\right)|000\rangle + \frac{1}{\sqrt{2}}|001\rangle + \frac{1}{\sqrt{10}}|011\rangle + \frac{i}{2}|111\rangle .$$

   (a) Is the state normalized? What is the probability that the system is found in the state $|000\rangle$ if all 3 qubits are measured?

   (b) What is the probability that a measurement on the first qubit only gives 0? What is the postmeasurement state of the system?

**(5+5)**

3. What is difference between classical bit and quantum bit (qubit). Suppose there is qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. What are the probabilities of finding 0 and 1 upon measurement. **(10)**

4. What do you mean by entanglement? Discuss all four types of entangled Bell state. Also draw the quantum circuit to generate any one of the Bell states. **(10)**

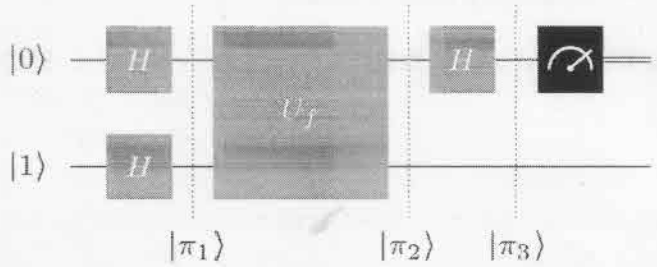
5. (a) Explain how BB84 quantum cryptography scheme secures the communication.

   (b) In B92 protocol, state the four rules based on which Bob decides his observation in + basis or × basis. **(5+5)**

6. What do you mean by superdense coding? Explain its complete working with proper circuit diagram. **(10)**

7. Find below the quantum circuit for Deutsch's algorithm. Derive the quantum state $|\pi_1\rangle$, $|\pi_2\rangle$, and $|\pi_3\rangle$.

**(10)**




8. Draw the circuit diagram of Shor's 9-qubit code for quantum error correction. Consider a Y error occurs at 1st qubit. Show the status of quantum state at each step. **(10)**

9. Given the search space $|000\rangle, |001\rangle, |010\rangle, |011\rangle, |100\rangle, |101\rangle, |110\rangle, |111\rangle$ (that is, we are given a function $f : \sum^3 \rightarrow \sum$ where $\sum = \{0, 1\}$). Suppose $|011\rangle$ is marked as present (in other words, $f(011) = 1$). Calculate the different steps of Grover's algorithm to find the element $|011\rangle$ in the search space. **(10)**

10. Describe any two of the following quantum postulates:

    1. State postulate
    2. Measurement postulate
    3. Evolution postulate
    4. Composition postulate

**(10)**